National University of Computer & Emerging Sciences, Karachi
Fall-2025 School of Computing (BSCS, BSSE, BSCY, BSAI)

Assignment # 3

Subject: Database Systems -CS2005 **Total Marks: 20**	**Post Date: 29/10/2025** **Due Date:** 9/11/2025
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Instructions to be strictly followed.

- For all questions involving SQL Queries:
  - ***Submit the SQL Scripts in a .txt file.***
- It should be obvious that submitting your work after the due date will result in zero points being awarded.
- Plagiarism (copying/cheating) and late submissions result in a zero mark.

Question#1 In 2025, the rise of **AI-powered cyberattacks, data breaches**, and **geopolitical cyber conflicts** has led the **Global Cyber Defense Alliance (GCDA)** to build a **centralized Cyber Threat Intelligence (CTI) database.** The GCDA database includes the following relations:

Relation	Attributes	Description
COUNTRY	(Country_ID, Country_Name, Region, GDP_Rank)	Stores country details
THREAT	(Threat_ID, Threat_Name, Category, Severity_Level)	Stores types of cyber threats (e.g., Ransomware, Phishing, AI Deepfake Attacks)
INCIDENT	(Incident_ID, Threat_ID, Country_ID, Date_Reported, Impact_Score)	Each reported cyber incident and its severity
RESPONSE_TEAM	(Team_ID, Country_ID, Team_Name, Specialization)	Cyber response teams in each country
ACTION_TAKEN	(Incident_ID, Team_ID, Action_Type, Resolution_Status)	Which team responded to which incident and what was done

You are a **Database Analyst** working for GCDA. Write the **Relational Algebra** expression and one line explanation of each command for the following:

1. List the names of countries that have reported critical-level threats (Severity_Level = "Critical") in 2025.
2. Retrieve the names of response teams that successfully resolved ransomware-related incidents.
3. Find all countries that have not yet formed any cyber response team.
4. Display the names of threats that have been reported by more than one country.
5. List all AI-based attack incidents (Category = "AI Attack") along with their country names and impact scores greater than 80.

<u>Question#2</u> Consider a database with the following schema:

Cafe (cafeID, cafeName, city, managerID)
Employee (empID, name, address, gender, salary, cafeID)
Customer (custID, name, address, phone)
Orders (orderID, custID, cafeID, totalAmount, orderDate)
MenuItem (itemID, itemName, category, price, isAvailable)
OrderDetails (orderID, itemID, quantity)
Supplier (supplierID, name, city, contactNo)
Supplies (supplierID, itemID, supplyDate, cost)

Describe the relations that would be produced by the following relational algebra operations (Textual meaning required) and also provide sql queries of each algebraic expression:

1. $\pi_{itemName}(\sigma_{isAvailable = 0 \wedge price > 500}(MenuItem))$
2. $\pi_{itemName}(\sigma_{(category = 'Beverage' \wedge price > 300) \vee category = 'Dessert'}(MenuItem))$
3. $\pi_{itemName,\ price}(\sigma_{category = 'Snack' \wedge isAvailable = 1}(MenuItem))$
4. $\pi_{itemName,\ price/100}(\sigma_{category = 'Snack' \wedge isAvailable = 1}(MenuItem))$
5. $\pi_{supplierID}(Supplies \bowtie MenuItem \bowtie \sigma_{itemName = 'Cappuccino'}(MenuItem))$
6. $\pi_{Employee.name,\ Cafe.cafeName}(\sigma_{Employee.cafeID = Cafe.cafeID \wedge Employee.address = Cafe.city}(Employee \bowtie Cafe))$
7. $\pi_{empID}(\sigma_{salary \neq 50000}(Employee))$
8. $\pi_{supplierID}(\sigma_{city = 'Karachi'}(Supplier)) \cap \pi_{supplierID}(\sigma_{city = 'Lahore'}(Supplier))$
9. $\pi_{itemName}(MenuItem) - \pi_{itemName}(MenuItem \bowtie Supplies)$